

## Exercise Sheet 7

Discrete Mathematics by Hongfei Fu, 2022.10.13

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1. Determine whether the relation  $R$  on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where  $(x, y) \in R$  if and only if
- c)  $x - y$  is a rational number.

**Answer Area:**

- $\therefore \forall x, x$  is a real number,  $x - x = 0$  is a rational number.  
 $\therefore \forall x, xRx$ . The relation  $R$  is reflexive.  
 $\therefore$  if  $x - y$  is a rational number, then  $y - x$  is a rational number.  
 $\therefore xRy \rightarrow yRx$ . The relation  $R$  is symmetric.  
 $\therefore (1, 2) \in R, (2, 1) \in R, 1 \neq 2$   
 $\therefore (xRy) \cap (yRx) \rightarrow x = y$  is False. The relation  $R$  is not antisymmetric.  
 $\therefore$  If  $x - y$  is a rational number,  $y - z$  is a rational number, then  $x - z = (x - y) - (y - z)$  is a rational number.  
 $\therefore (xRy) \cap (yRz) \rightarrow (xRz)$ . The relation  $R$  is transitive.

2. Determine whether the relation  $R$  on the set of all integers is reflexive, symmetric, antisymmetric, and/or transitive, where  $(x, y) \in R$  if and only if
- a)  $x \neq y$ .  
b)  $xy \geq 1$   
f)  $x$  and  $y$  are both negative or both nonnegative.  
h)  $x \geq y^2$

**Answer Area:**

- a) symmetric  
b) symmetric, transitive  
f) reflexive, symmetric, transitive.  
h) antisymmetric, transitive.

3. Exercises 36 deal with these relations on the set of real numbers:

$R_1 = \{(a, b) \in \mathbf{R}^2 \mid a > b\}$ , the "greater than" relation,  
 $R_2 = \{(a, b) \in \mathbf{R}^2 \mid a \geq b\}$ , the "greater than or equal to" relation,  
 $R_3 = \{(a, b) \in \mathbf{R}^2 \mid a < b\}$ , the "less than" relation,  
 $R_4 = \{(a, b) \in \mathbf{R}^2 \mid a \leq b\}$ , the "less than or equal to" relation  
 $R_5 = \{(a, b) \in \mathbf{R}^2 \mid a = b\}$ , the "equal to" relation,  
 $R_6 = \{(a, b) \in \mathbf{R}^2 \mid a \neq b\}$ , the "unequal to" relation.

Find:

- c)  $R_1 \circ R_3$   
e)  $R_1 \circ R_5$   
f)  $R_1 \circ R_6$   
h)  $R_3 \circ R_3$

**Note:** Please give reasons for your answers in c) and h)

**Answer Area:**

c)  $R_1 \circ R_3 = \mathbf{R}^2$

The proof is as follows:

- If  $a, b \in R_1 \circ R_3$  Then  $\exists c \in \mathbf{R}, aR_3c \cap cR_1b$ .  
Then  $a \in \mathbf{R}$  and  $b \in \mathbf{R}, a, b \in \mathbf{R}^2$
- If  $a, b \in \mathbf{R}^2$  Then  $\exists c \in \mathbf{R}, c > a$  and  $c > b$   
Then  $\exists c, aR_3c$  and  $cR_1b$   
Then  $a, b \in R_1 \circ R_3$

e)  $R_1 \circ R_5 = \{(a, b) \in \mathbf{R}^2 | a > b\},$

f)  $R_1 \circ R_6 = \mathbf{R}^2$

h)  $R_3 \circ R_3 = \{(a, b) \in \mathbf{R}^2 | a < b\}$

The proof is as follows:

- If  $a, b \in R_3 \circ R_3$   
Then  $\exists c \in R, aR_3c \cap cR_3b$ .  
Then  $a < c$  and  $c < b$ .  
Then  $a < c < b, aR_3b$   
Then  $a, b \in R_3$
- If  $a, b \in R_3$ .  
Then  $aR_3b, a < b$ .  
Then  $\exists c \in R, c = \frac{a+b}{2}, a < c < b$ .  
Then  $aR_3c \cap cR_3b$ .  
Then  $a, b \in R_3 \circ R_3$

4. Prove that for all relations  $R$  and  $S$ ,  $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$ .**Answer Area:**

- Suppose that  $x, y \in (R \circ S)^{-1}$ .  
 $(x, y) \in (R \circ S)^{-1}$ .  
 $\therefore (y, x) \in R \circ S$   
 $\therefore \exists z, ySz \cap zRx$   
 $\therefore zR^{-1}y \cap xR^{-1}z$ .  
 $\therefore (x, y) \in S^{-1} \circ R^{-1}$ .
- Suppose that  $(x, y) \in S^{-1} \circ R^{-1}$ .  
 $\therefore (x, y) \in S^{-1} \circ R^{-1}$ .  
 $\therefore \exists z, xR^{-1}z \cap zS^{-1}y$ .  
 $\therefore zRx \cap ySz$ .  
 $\therefore (y, x) \in R \circ S$ .  
 $\therefore (x, y) \in (R \circ S)^{-1}$

5. Prove that the composition operator  $\circ$  is associative over relations.**Answer Area:**The proof of  $(R \circ S) \circ T = R \circ (S \circ T)$  is as follows:

- Suppose that  $(x, y) \in (R \circ S) \circ T$ .  
 $\therefore (x, y) \in (R \circ S) \circ T$ .  
 $\therefore \exists z, xTz \cap z(R \circ S)y$ .  
 $\therefore \exists t, zSt \cap tRy$ .  
 $\therefore x(S \circ T)t \cap tRy$ .  
 $\therefore (x, y) \in R \circ (S \circ T)$
- Suppose that  $(x, y) \in R \circ (S \circ T)$ .  
 $\therefore (x, y) \in R \circ (S \circ T)$ .  
 $\therefore \exists z, (x, z) \in S \circ T \cap (z, y) \in R$ .  
 $\therefore \exists t, xTt, tSz$  and  $zRy$ .  
 $\therefore xTt, t(R \circ S)y$ .  
 $\therefore (x, y) \in (R \circ S) \circ T$ .

In conclusion,  $(R \circ S) \circ T = R \circ (S \circ T)$ . The composition operator  $\circ$  is associative over relations.

6. Prove that a relation  $R$  on a set  $A$  is transitive iff  $R \circ R \subseteq R$ .

**Answer Area:**

- Suppose that relation  $R$  on a set  $A$  is transitive.  $\therefore \forall x, y \in A, (x, y) \in R \circ R, \exists z \in A, xRz \cap zRy$ .  
 $\therefore R$  is transitive.  
 $\therefore xRy, (x, y) \in R$ .  
 $\therefore R \circ R \subseteq R$ .
- Suppose that  $R \circ R \subseteq R$ .  $\therefore R \circ R \subseteq R$ .  
 $\therefore \forall (x, y) \in R \circ R, (x, y) \in R$ .  
 $\therefore \forall x, y, z \in A (xRz \cap zRy), (x, y) \in R \circ R$ , then  $(x, y) \in R$ .  
 $\therefore \forall x, z, y \in A (xRz \cap zRy \rightarrow xRy)$ .  
 $\therefore$  relation  $R$  on a set  $A$  is transitive.

$\therefore$  A relation  $R$  on a set  $A$  is transitive iff  $R \circ R \subseteq R$ .

7. Prove that a relation  $R$  on a set  $A$  is antisymmetric iff  $R \cap R^{-1} \subseteq I_A$ , where  $I_A := \{(a, a) \mid a \in A\}$ .

**Answer Area:**

- Suppose that relation  $R$  on a set  $A$  is antisymmetric.  $\therefore \forall x, y \in A, (x, y) \in R \cap R^{-1}, xRy \cap xR^{-1}y$ .  $\therefore xRy \cap yRx$ .  
 $\therefore R$  is antisymmetric.  
 $\therefore x = y, (x, y) \in I_A$ .  
 $\therefore R \cap R^{-1} \subseteq I_A$ .
- Suppose that  $R \cap R^{-1} \subseteq I_A$ .  $\therefore R \cap R^{-1} \subseteq I_A$ .  
 $\therefore \forall x, y \in A, (x, y) \in R \cap R^{-1}, (x, y) \in I_A$ .  
 $\therefore \forall x, y \in A (xRy \cap xR^{-1}y), (x, y) \in I_A$ .  
 $\therefore \forall x, y \in A (xRy \cap yRx), x = y$ .  
 $\therefore \forall x, y \in A ((xRy \cap yRx) \rightarrow (x = y))$ .  $\therefore$  relation  $R$  on a set  $A$  is antisymmetric.

$\therefore$  A relation  $R$  on a set  $A$  is antisymmetric iff  $R \cap R^{-1} \subseteq I_A$ , where  $I_A := \{(a, a) \mid a \in A\}$ .

8. Prove or disprove that for any binary relations  $R, S, T$  on a nonempty set  $A$ , we have that  $(R \circ S) \cap T = \emptyset$  iff  $(R^{-1} \circ T) \cap S = \emptyset$ .

**Answer Area:**

- Suppose that  $(R \circ S) \cap T \neq \emptyset$ .  $\therefore (R \circ S) \cap T \neq \emptyset$ .  
 $\therefore \exists x, y \in A, (x, y) \in R \circ S$  and  $xTy$ .  
 $\therefore \exists z \in A, xSz \cap zRy, xTy$ .  
 $\therefore xSz, xTy \cap yR^{-1}z$ .  
 $\therefore (x, z) \in S$  and  $(x, z) \in R^{-1} \circ T$ .  
 $\therefore (x, z) \in (R^{-1} \circ T) \cap S$ .  
 $\therefore (R^{-1} \circ T) \cap S \neq \emptyset$ .  
 $\therefore (R^{-1} \circ T) \cap S = \emptyset \rightarrow (R \circ S) \cap T = \emptyset$
- Suppose that  $(R^{-1} \circ T) \cap S \neq \emptyset$ .  $\therefore (R^{-1} \circ T) \cap S \neq \emptyset$   
 $\therefore \exists x, y \in A, (x, y) \in R^{-1} \circ T$  and  $xSy$ .  
 $\therefore \exists z \in A, xTz \cap zR^{-1}y$ .  
 $\therefore xSy \cap yRz, xTz$ .  
 $\therefore (x, z) \in R \circ S$  and  $(x, z) \in T$ .  
 $\therefore (R \circ S) \cap T \neq \emptyset$ .  
 $\therefore (R \circ S) \cap T = \emptyset \rightarrow (R^{-1} \circ T) \cap S = \emptyset$ .

In conclusion,  $(R \circ S) \cap T = \emptyset$  iff  $(R^{-1} \circ T) \cap S = \emptyset$ .